Inverted MAML Card 1: Round 15 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.055, L2=1920, Mom=0.85, Clamp=±90 LSB

1. Round 15 Summary & Cumulative Comparison (R1–R14)

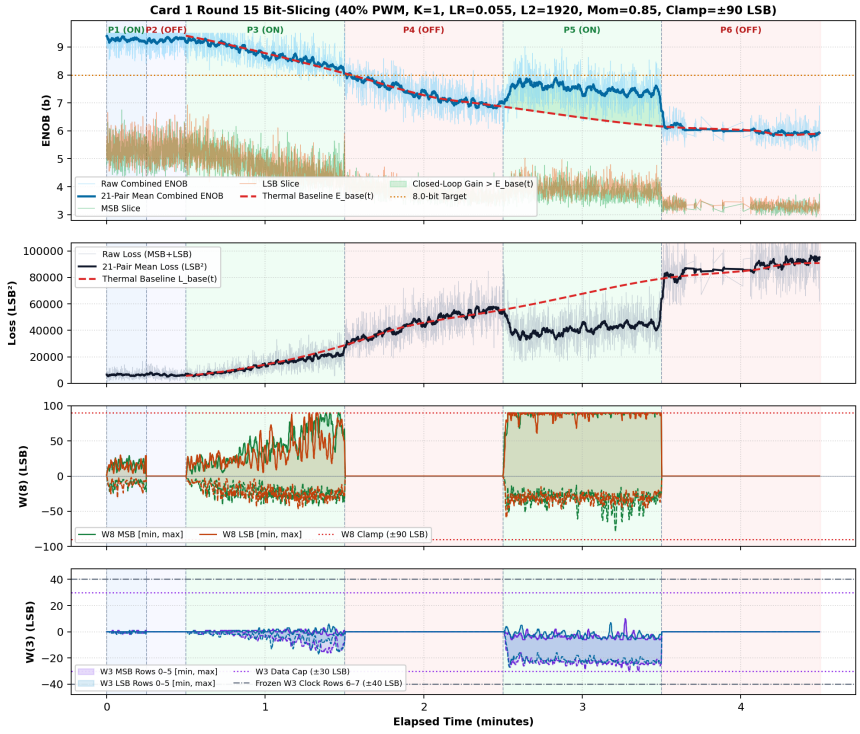
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 15 Detrended Trend Metrics:** Lift vs E_base = **+0.78 b**, Loss Ratio = **1.55x**, Climb = **+0.97 b**, Jitter = **0.39 b**, Recovery = **0.67**, Score = **+2.117**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R7dc_fix*	0.055	2048	0.85	±75	+1.04b	1.74x	+1.17b	0.40b	0.50	+2.665
R8dc_fix	0.055	2048	0.85	±75	+1.16b	1.85x	+0.68b	0.42b	0.00	+1.855
R9dc_fix	0.055	1920	0.85	±75	+0.81b	1.59x	+0.66b	0.38b	0.50	+1.756
R10dc_fix	0.055	1920	0.85	±78	+0.96b	1.67x	+1.13b	0.40b	0.00	+1.898
R11dc_fix	0.055	1920	0.85	±80	+0.95b	1.70x	+0.98b	0.42b	0.00	+1.810
R12dc_fix	0.055	1920	0.85	±82	+1.04b	1.75x	+0.83b	0.42b	0.00	+1.721
R13dc_fix	0.055	1920	0.85	±85	+1.12b	1.78x	+1.06b	0.43b	0.50	+2.645
R14dc_fix	0.055	1920	0.85	±90	+0.78b	1.55x	+0.97b	0.39b	0.67	+2.117

2. Phase-by-Phase Telemetry (3,886 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	325	9.31->9.25 (9.26b)	-	5510->6163	[29,31]	2	325
P2	Cold / Regler OFF	299	9.21->9.30 (9.25b)	-	6711->5958	[0,0]	0	0
P3	Warm / Regler ON	1051	9.15->8.44 (8.78b)	+0.11 b	7039->19327	[90,90]	17	1051
P4	Warm / Regler OFF	923	7.85->6.99 (7.36b)	-0.01 b	32450->53454	[0,0]	0	0
P5	Warm / Regler ON	803	7.12->7.38 (7.50b)	+1.13 b	51142->43884	[90,90]	30	803
P6	Warm / Regler OFF	485	6.14->5.90 (5.99b)	+0.03 b	81824->92709	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 15

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±90 LSB, while W(3) rows 0–5 stay within ±30 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 15 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

